

Ian Hsu

Software Engineer

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I am a Master of IT student at the University of Auckland and a former Software Engineer at Acer with three years of experience. I specialize in AWS cloud solutions, designing event-driven services, web services, RESTful APIs, and CI/CD workflows. As a self-taught developer, I thrive on learning quickly and adapting to complex technical environments.

Here is my video introduction: <https://app.previewme.com/p/jv66he>

Work Experience

● Software Developer Intern | DXR Limited (NZ Limited Company) Nov 2025 - Mar 2026

- **Led a team** of three developer interns to engineer a **full-stack** music event platform using **AWS serverless architecture**.
- Music Event Platform
 - Developed backend **RESTful APIs** using AWS Lambda, following clean architecture principles.
 - Built an **event-driven media pipeline** using AWS S3 event triggers to automatically generate image thumbnails after file uploads.
 - Built an admin dashboard for platform management using **React.js** and Material-UI.
- CI/CD and Infrastructure Automation
 - Set up CI/CD pipelines using **GitHub Actions** and **AWS Amplify Hosting** for the public frontend, admin frontend, and backend projects.
 - Used **Terraform** to build and manage the cloud infrastructure across multiple stacks.

● Software Engineer | Acer Inc. Aug 2022 - Jul 2025

- Managed both Developer and PM roles, **translating technical concepts to non-technical stakeholders** to draft clear and actionable technical specs.
- Vulnerability Log Automation & Reporting System
 - **Identified maintenance bottlenecks** (technical debt) and built an event-driven system (**AWS Lambda, SQS**) to automate vulnerability reporting.
 - Reduced manual effort by **~98%** (from 5 hours to 5 minutes).
- Self-Service Data Analytics Platform
 - Resolved the issue of scattered resources across multiple departments by building a platform for better data management and reporting.
 - Developed a backend service using **NestJS**, deployed on AWS ECS Fargate.
 - Processed large CSV files using data streaming and stored the data in **Elasticsearch**.
 - Built an **ETL** data pipeline with **AWS Glue** to extract and clean data.
 - Added reliable logging and graceful shutdown mechanism to backend service.
- CI/CD and Infrastructure Automation
 - Designed and implemented CI/CD pipelines in **GitLab CI/CD** to speed up deployment.
 - Utilised **Terraform** and **AWS CDK** to write Infrastructure as Code (IaC) and manage cloud resources.
- System Monitoring
 - Streamlined incident response by implementing an automated **error monitoring system** for ~100 AWS Lambda functions and RDS using **CloudWatch** and **Grafana**.
 - Integrated with MS Teams for **real-time alerting** to reduce system downtime.
- Data Pipeline and Infrastructure Migration
 - Migrated data pipeline service from self-managed Kubernetes to AWS ECS Fargate to improve system scalability.
 - Collected and forwarded system logs for network devices using **Fluentd**.
 - Developed a custom Grafana web application using **React.js** and **Golang**.

Certifications

- **AWS Certified Solutions Architect - Associate**

Jul 2023

Education

- **University of Auckland** | *Master of Information Technology* *Jul 2025 - Present*
- **National Kaohsiung University of Hospitality and Tourism** | *Bachelor of Restaurant Management* *Sep 2017 - Jun 2021*

Software Projects

- **aws-lambda-events (NPM open-source package)** *Dec 2025 - Present*
 - Created an open-source npm package that standardise AWS API Gateway event inputs (REST v1.0 and HTTP v2.0) for Lambda handlers.
 - Engineered wrappers that unify the retrieval of path parameters, query strings, and body inputs across multiple payload versions.
 - Build a validator adapter pattern supporting Joi and Zod, with a consistent ValidationError interface to decouple business logic from parsing boilerplate.
 - Resolved complex edge cases, including base64-encoded bodies, duplicate query parameters, and case-insensitive header lookups through automated handling.
 - **Skills used:** TypeScript, Node.js, AWS Lambda, API Gateway, Joi, Zod, Jest.
 - **Link:** <https://www.npmjs.com/package/aws-lambda-events>
- **aws-apigw-jwt-auth (NPM open-source package)** *Nov 2025 - Present*
 - Created an open-source npm package that simplifies working with JWT authorizers behind AWS API Gateway and Lambda.
 - Designed reusable authorization utilities (e.g. checking user groups/scopes) to keep business logic handlers small and focused.
 - Consolidated redundant authentication logic and streamline error handling.
 - **Skills used:** TypeScript, Node.js, AWS Lambda, API Gateway, JWT, Jest.
 - **Link:** <https://www.npmjs.com/package/aws-apigw-jwt-auth>
- **Forum website** *2021 - 2022*
 - Implement third-party registration and login mechanisms (Google and Meta).
 - Develop functionality for publishing articles and media content.
 - Implement keyword-based search functionality for articles.
 - Develop the "Click to Like" feature for articles and comments.
 - **Skills used:** React.js, Node.js (Express), Docker, MongoDB (Mongoose), AWS, JWT.
 - **Link:** <https://github.com/ianhsu0116/Ocard>

Technical Skills

- **Languages & Frameworks:** TypeScript, Node.js, C# (.NET), React.js, React Native, MySQL, PostgreSQL, MongoDB, Elasticsearch.
- **Infrastructure & DevOps:** AWS, Azure, Terraform, CDK, Docker, Fluentd, CI/CD (GitHub, GitLab), Unit-test, Linux.

Professional Skills

- **Agile:** Scrum, Kanban, Sprint Planning, Retrospectives.
- **Optimization:** Workflow Automation, Bottleneck Analysis, Process Improvement.
- **Communication:** Technical Specs, Dev/PM Coordination, Stakeholder Management.
- **Collaboration:** Self-learning, Brainstorming, Feedback Loops.

Hackathon & Volunteering

- **2026 WEB3UOA Hackathon**
- **Kayak Clean Up Volunteer** - Mountains to Sea Conservation Trust